Thamill Indian B U
Cohort-15 (AI Engineer)

Day 07 Post-Session Assignment & Engineering Reflection

AI Engineering Program | Day 07

Part 1 - Build evidence

- **Working End-to-End RAG Pipeline**: Successfully connected the Day 06 vector retriever and structural chunks to a local LLM generation layer with strict context construction.
- **Captured Execution Results (Sample JSON Output for Question 1)**:


```
{
  "question": "How many vacation days do employees receive each year?",
  "retrieved_chunks": [
    {
      "rank": 1,
      "chunk_id": 1,
      "score": 0.4911,
      "text": "Employees receive 24 days of paid annual leave per calendar year.
      "metadata": {"section": "Internal Operations Handbook > Human Resources >
    }
  ],
  "answer": "Internal Operations Handbook > Human Resources > Annual Leave: Empl
  "citations": ["Source ID: 1"],
  "evidence": [{
    "claim": "Employees receive 24 days of paid annual leave per calendar year.'
    "chunk_id": 1,
    "quote": "Employees receive 24 days of paid annual leave per calendar year.'
  }],
  "status": "answered"
}
```


- **Unsupported Question Handling (Question 6 - Parental Leave)**: When tested with *"What is the company's parental leave duration?"*, the system successfully bypassed model memory hallucinations and returned an explicit insufficient-evidence response: *"The available evidence does not provide enough information regarding parental leave duration."*

Part 2 - Quality observations

- **Strongest Answer**: Question 1 (*Vacation days*) produced the strongest grounded answer because the exact lexical match and high semantic similarity score correctly surfaced Chunk 1 right at Rank 1.
- **Weakest Result**: Complex policy comparison queries (e.g., Question 7) showed occasional weakness where the retriever split relevant details across multiple chunks, requiring robust context assembly.
- **Hallucination Check**: No generated answer contained claims outside the retrieved context blocks.
- **Citation Mapping**: Every citation mapped directly to a valid supplied chunk ID, ensuring zero fabricated sources.

Thamill Indian B U
Cohort-15 (AI Engineer)

Part 3 - Failure analysis

- **Failure Case**: During multi-hop policy reasoning (Question 7), the retriever occasionally surfaced only one of the two requested policy sections within the default Top-2 limit.
- **Root-Cause Classification**: Classified as a **Retrieval & Context Construction** failure (insufficient Top-K breadth rather than model hallucination).
- **Supporting Evidence**: Inspection of retrieved chunks showed that relevant secondary sections were ranked just outside the top results (Rank 3/4).
- **Change Tested**: Increased Top-K from 2 to 3, which successfully captured both target policy chunks in the prompt context.

Part 4 - Engineering reasoning

- **Why a good prompt cannot fully fix bad retrieval**: If the retriever fails to surface the correct document chunk, the LLM has zero context to read from. A prompt can enforce formatting and guardrails, but it cannot invent missing facts.
- **Why citations are useful but not proof of correctness**: Citations prove *where* the model looked, but they do not guarantee that the model interpreted the source text correctly or avoided misrepresenting its contents.
- **Why unsupported questions must be tested deliberately**: To verify that system guardrails and prompts successfully block hallucinations and trigger safe insufficient-evidence fallback messages instead of guessing.
- **What happens if Top-K is increased significantly**: Injecting too many chunks expands the context window, increasing latency, token cost, and the risk of "lost-in-the-middle" attention degradation where the model ignores critical facts.

Part 5 - FDE reflection

- **Stricter Business Questions**: Financial compliance guidelines, legal indemnities, and medical/safety procedures require absolute factual adherence with zero tolerance for ambiguity.
- **Conflict/Missing Evidence**: The system must fail safely by refusing to answer, logging an audit event, or escalating to a human review queue.
- **Trust Model Explanation**: Explain to customers that the assistant acts as a strictly bounded search and synthesis tool that quotes certified company documents rather than relying on unverified internet memory.
- **Pilot Readiness Evidence**: Demonstrate high precision/recall metrics on a curated test set, zero hallucination rates on unsupported test queries, and verifiable source citations.